Samaneh Shirinnezhad

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Education

B.Sc. in Computer Engineering *September 2012 – December 2016*
Jundi Shapur University of Technology Dezful, Iran

- **Capstone Topic:** Design and Development of a VR Game with Motion Sensor Integration for Mobile Platforms using Unity and Google Cardboard
- **GPA:** 145 credits program with GPA of 17.51/20. GPA of the last two years is (3.81 / 4.00)
- **Selected Courses:** Advanced Programming, Algorithm Design, Data Structures, Artificial Intelligence, Data Transmission, Discrete Structures, Internet Engineering

Research Interests

- Human-Computer Interaction
- User-Interface Design
- User-Experience Design
- Information Visualization
- Human-AI Collaboration
- AI in Education

Skills

Languages: English (IELTS Academic – Overall Band Score: 8.5), Persian (Native)
Programming Languages: C++, C#, Java, Python, R, Julia, SQL
Data Science & Machine Learning: NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, Keras, PyTorch, Gensim, NLTK, SpaCy, Transformers
Web Development: HTML, CSS, JavaScript, TypeScript, React, Next.js, Node.js (Express), PostgreSQL
Dev & Research Tools: Jupyter Notebook, RStudio, LaTeX/Overleaf, Tableau, Excel, NVivo, Git, Docker
Design Tools: Figma, Adobe Creative Suite (Photoshop, Illustrator, Premiere)
Virtual & Augmented Reality: Unity3D, VR SDKs

Work Experience

Bartar Language School *2016 – Present*
English Instructor & Digital Learning Support Contributor Dezful, Iran

- Taught English as a Second Language (ESL) courses across multiple proficiency levels, using interactive activities to strengthen communication skills and learner confidence.
- Designed user interface components for Bartar's Android language-learning app, covering course navigation, lesson exercises, vocabulary cards, class schedules, and student profiles.
- Connected classroom needs with digital learning features, including assignment flow, student progress visibility, and teacher-student communication.
- Supported teacher-training activities through lesson review, mock-class observation, and feedback for new instructors.

Freelance Work [ ] *2018 – Present*
Data Scientist & Frontend/UI Developer Remote

- Delivered data-driven web tools, dashboards, and visual reporting systems for clients across business intelligence, GIS, and social media analytics projects.
- Designed responsive interfaces that made complex data easier to navigate, compare, and understand for non-technical users.
- Built data-cleaning workflows, machine learning models, and visualization pipelines to support analysis, forecasting, and client decision-making.
- Applied frontend development, UI design, and data visualization skills to improve the clarity and usability of interactive tools.

Jundi Shapur University of Technology *Fall 2016*
Teaching Assistant — Data Structures and Algorithms Dezful, Iran

- Assisted in teaching undergraduate students core concepts in algorithms and data structures.
- Graded assignments, midterms, and final exams; held tutorials and provided 1-on-1 academic support.

Islamic Republic of Iran Railways *June 2016 - September 2016*
Undergraduate Intern Andimkeshk, Iran

- Collaborated on a research initiative to apply VR to industrial training for transportation maintenance simulations.
- Built early prototypes and test scripts; addressed technical and usability issues with cross-disciplinary engineering/design teams.
- Contributed to running pre- & post-skill assessments and analyzed training-time logs to evaluate learning outcomes.
- Findings: 40% lower training costs and +25% skill-assessment scores vs. baseline modules.
- Documented results and iteration notes, establishing a simple metrics pipeline for future training evaluations.

Research Experience

Bartar Language School [🌐] *2025 – 2026*
Teacher-in-the-Loop AI Feedback Design Study Dezful, Iran

- Led a pilot HCI study with 8 English/ESL teachers to examine how AI-assisted writing feedback should be reviewed, edited, and controlled before reaching learners.
- Designed four prototype feedback workflows: direct correction, guided hints, teacher-controlled feedback, and staged feedback.
- Collected teacher ratings, comparison responses, think-aloud notes, and co-design feedback to analyze trust, classroom fit, student effort, and copying risk.
- Found that teachers preferred feedback designs that kept AI under teacher supervision, with approval, editing, and staged student revision before full correction.
- Developed a Teacher-in-the-Loop Feedback Control Framework to guide future AI-supported writing feedback tools at Bartar.

Research Collaboration with Davoud Ghahremanlou *2023 – 2024*
Bibliometric Analysis, Machine Learning & Research Visualization Iran / Remote

- Analyzed renewable energy literature using bibliometric and machine learning methods to identify trends, topic clusters, and research gaps.
- Developed data-processing and visualization workflows to make large-scale research patterns easier to interpret.
- Connected technical analysis with user-facing communication by turning complex model outputs into clear visual and written explanations.
- Led the research that resulted in a peer-reviewed publication through contributions to conceptualization, methodology, investigation, programming, visualization, original draft writing, and editing (see Publications section).

Undergraduate Research Assistant to Dr. Mohsen Shakiba *2015 – 2016*
Virtual Reality Lab at Jundi Shapur University of Technology Dezful, Iran

- Assisted in research on applying VR technologies to mobile platforms using Unity and Google Cardboard, forming the foundation of my undergraduate capstone project.
- Implemented motion-sensor integration techniques to explore immersive interaction and usability in resource-constrained environments.
- Contributed to experimental design and early testing of VR gameplay mechanics, linking research insights to practical development.

Recent Publications

J=Journal, C=Conference, P=In Press, S=Submitted, U=Under Review

Find me at: Google Scholar ORCID

[J.1] **Navigating the Canadian Renewable Energy Landscape through Bibliometric and Machine Learning Insights**.
Samaneh Shirinnezhad, & Davoud Ghahremanlou.
International Journal of Global Warming, Vol. 37, No. 1, 2025. [DOI]

Selected Projects

Teacher-in-the-Loop AI Feedback Prototype *2025 – 2026*
Tools: React, JavaScript, OpenAI API, UX Prototyping [🌐]

- Built an interactive prototype for AI-assisted English writing feedback where teachers review, edit, approve, or delay feedback before students see it.
- Implemented multiple feedback workflows, including direct correction, guided hints, teacher-controlled feedback, and staged feedback after student revision.
- Used the prototype in a Bartar pilot study with English/ESL teachers to compare teacher control, student effort, trust, classroom fit, and copying risk.
- Translated pilot findings into interface requirements for teacher approval, editable feedback, hint-first support, and student revision before full correction.

Bartar Learning App Redesign *2025-2026*
Tools: Figma, UX/UI Design, Interaction Design, Prototyping [🌐]

- Led a redesign concept for Bartar's language-learning app, extending the original mobile experience into a student-and-teacher platform for guided language practice.
- Designed core screens for daily learner check-ins, smart translation, accessibility controls, class support, and a teacher dashboard.
- Focused on reducing passive copying by adding try-first interactions, hint-based support, teacher visibility, and clearer feedback flows.
- Prepared annotated screens and a product brief to explain the design logic, user needs, and learning-support features.

SceneSketch: AI-Powered Reading Companion *October 2025*
Tools: React, Tailwind, epub.js, Node/Express, OpenAI, Stability AI [GitHub]

- Built an EPUB reader that lets users highlight a passage and generate a style-aware visual scene to support reading and imagination.
- Implemented facing-page reading, page navigation, bookmarks, highlights, saved prompts, and an inline scene-generation workflow.
- Explored how text-to-image support can help readers connect written passages with visual context, especially for second-language learners and readers who benefit from multimodal support.

AirSpell: Air-Writing for Early Literacy *July 2025*
Tools: React, TensorFlow.js, HTML5 Canvas, JavaScript [GitHub]

- Developed a webcam-based spelling practice tool that lets early learners form letters in the air using hand movement.
- Combined motion-based interaction with visual feedback to support letter recognition, spelling practice, and learner engagement.
- Designed controls for practice flow, correction, and repeated attempts to make the activity usable for young learners.

SmartTextEnhancer *May 2025*
Tools: JavaScript, Chrome Extensions, OpenAI API [GitHub]

- Built an AI-powered Chrome extension for simplifying, translating, and rephrasing text directly on webpages.
- Added in-place text transformation so users can adapt difficult content without switching tabs or copying text into another tool.
- Included reading-support options such as font-size adjustment and reading-friendly display controls.

Scholarly Snap Assistant *2025*
Tools: HTML, Tailwind CSS, JavaScript, Node.js, Express, AssemblyAI, OpenAI API [GitHub]

- Built a web app for recording or uploading lecture and meeting audio, generating transcripts, and producing AI-assisted summaries.
- Implemented summary-length controls and export options for transcripts and summaries in TXT, PDF, and DOC formats.
- Designed a responsive interface to help students and researchers review spoken content more efficiently.

Insights into ChatGPT Research
May 2023

Tools: Python, BeautifulSoup, Google Scholar API, Pandas, NLTK, spaCy, LDA

- Analyzed nearly 1,000 research papers on ChatGPT using exploratory text analysis and LDA topic modeling.
- Mapped dominant research themes, including education, applications, ethics, scientific writing, and human–AI interaction.
- Identified underexplored areas related to trust, adoption, and how people understand AI tools in practical settings.

Bartar English App: Original Android UI Design
2016 – 2017

Tools: Android UI, Material Design, Mobile Interface Design

- Designed UI patterns and screen layouts for Bartar's first Android language-learning app, including sign-in, course navigation, lessons, vocabulary cards, class schedule, and student profile screens.
- Created a simple Material Design interface for students to continue learning between classes through lessons, word practice, progress tracking, and class information.
- Brought a classroom perspective into interface decisions, keeping the app approachable for learners across different ages and English levels.

Honors and Awards

Dean's List
Multiple Semesters (Fall 2014, Winter 2015, Fall 2015, Winter 2016)

Jundi Shapur University of Technology

- Recognized for consistent high academic performance over multiple semesters.
- Demonstrated exceptional academic dedication and achievement, placing in the top 2% of the class.

Annual JSU Programming Contest Winner
September 2015

Jundi Shapur University of Technology

- Secured first place in the annual JSU Programming Contest, specializing in algorithm design and implementation using C++.
- Excelled in solving complex problems under rigorous time constraints, demonstrating advanced proficiency in C++.
- Outperformed other competitors, showcasing superior competitive programming skills.
- Received a certificate and a recognition plaque for outstanding performance.

Service and Community Engagement

Community STEM Outreach
2023

Workshop Facilitator — Dezful, Iran

- Delivered introductory programming and problem-solving workshops for local high school students.
- Promoted early interest in computer science by mentoring participants through hands-on coding activities.

Local Public Library
2023

Volunteer Technology Mentor — Dezful, Iran

- Assisted older adults in developing computer literacy skills, including using email, online resources, and mobile applications.
- Organized small group sessions to promote digital confidence and independence, bridging generational gaps in technology use.

Jundi Shapur University of Technology
2015

Conference Organizer — Student Research Day — Dezful, Iran

- Organized a student-led academic event where undergraduate projects were showcased to faculty and peers.
- Coordinated call for submissions, scheduling, and volunteer teams, encouraging knowledge exchange across departments.